

Swiggy India Restaurant Analytics

Data-Driven Investment Insights for the Indian Food Delivery Market

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Sector

Food Technology / Restaurant Analytics

Dataset

<https://www.kaggle.com/datasets/rrkcoder/swiggy-restaurants-dataset/data>

Tools

Python, Pandas, Tableau Public

1 Executive Summary

This project analyzes 9,388 restaurant listings across six major Indian cities to identify key drivers of customer ratings and market opportunities.

Key Metrics:

- Average Rating: 3.996
- Average Price for Two: Rs 305
- Rated Restaurants: 76.7%

Core Insight Summary

Restaurant success on Swiggy is primarily driven by strategic positioning rather than pricing or promotions. Key factors such as cuisine type, city-level demand, and veg preference have a significantly stronger impact on customer ratings. The most attractive opportunities exist in segments where demand is high but quality remains underdeveloped.

Key Supporting Insights

- Cuisine is the strongest driver of performance, with categories like North Indian and Beverages consistently achieving higher ratings across cities.
- City dynamics play a critical role, as markets like Bangalore and Mumbai show both high demand and strong quality benchmarks, while others present growth opportunities.
- Pricing has a weak correlation with ratings.
- Promotional offers have minimal impact on ratings.
- Pure veg restaurants tend to perform better than non-veg counterparts.
- Demand-supply gaps exist across specific cuisine-city combinations.
- High-demand but low-quality segments represent promising entry points.
- Successful strategy requires alignment of cuisine, location, and positioning.

2 Sector Context & Problem Statement

India's food delivery market has experienced rapid growth due to urbanization, increasing digital adoption, and changing consumer lifestyles.

Customer ratings act as a key performance indicator influencing visibility and success.

Problem Statement

The objective is to determine key factors influencing restaurant performance and identify high-potential opportunities.

Key Questions

- Which cuisines and cities achieve higher ratings?
- Does pricing or discounting impact satisfaction?
- How does veg vs non-veg affect performance?
- Where do demand-supply mismatches exist?
- Which cuisine-city combinations offer best opportunities?

3 Data Engineering

Source: <https://www.kaggle.com/datasets/rrkcoder/swiggy-restaurants-dataset/data>

Dataset Overview

- Total Rows: 10,083
- Rated Restaurants: 7,738
- Cities: 6
- Cuisine Types: 80+
- Price Range: Rs 20 – Rs 2000

Data Cleaning & Preprocessing

- Extracted numeric ratings
- Standardized price column
- Converted rating counts
- Removed corrupted fields
- Filtered missing values

Feature Engineering

- Price Band
- Rating Tier
- Demand Tier
- Value Index
- Primary Cuisine

4 KPI Framework

- Average Rating: 3.996
- Total Restaurants: 7,432
- Average Price: Rs 305
- Rated Restaurants: 75.3%

5 Exploratory Data Analysis

Cuisine Insights

- South Indian: Highest rating (4.10)
- Biryani: Lowest rating (3.83)

City Insights

- Bangalore: Highest rating (4.13)
- Kolkata: Lowest rating (3.96)

Price Insights

- Mid-Premium (Rs 300–500) performs best
- Weak correlation between price and rating

6 Statistical Analysis

Cuisine Impact

- Strong influence on ratings
- North Indian, Beverages perform best

Price vs Rating

- Weak correlation ($r \approx 0.1$)

Promotions

- No significant impact

Veg vs Non-Veg

- Veg performs better

Summary

- Cuisine = strong driver
- Price = low impact
- Offers = minimal effect

7 Dashboard Design

Tableau Link:

https://public.tableau.com/app/profile/pratiti.paul/viz/SwiggyAnalysis_17773931032380
ExecutiveDashboard

7.1 Executive Performance Overview

KPIs:

- Avg Rating: 3.996
- Total Restaurants: 7,432
- Avg Price: Rs 305
- Rated: 75.3%

7.2 Operational Drill-Down

- Weak price-rating correlation
- Bangalore highest demand
- Heatmaps reveal performance gaps

7.3 Market Insights Dashboard

- Snacks and Beverages = strong value
- Discounts = minimal impact
- Veg performs better

8 Business Recommendations

- Focus on high-performing cuisines
- Target Bangalore, Hyderabad
- Use mid-premium pricing (Rs 300–500)
- Leverage veg positioning
- Exploit demand-quality gaps

9 Impact, Limitations & Future Scope

Impact

- Revenue optimization
- Better market positioning
- Cost-efficient expansion
- Reduced dependency on discounts

Limitations

- Snapshot data
- Missing ratings (23%)

Future Scope

- Time-series analysis
- Predictive models
- Expand to tier-2/3 cities

10 Contribution Matrix

Team Member	Dataset	Cleaning	KPI & Analysis	Dashboard	Report Writing	PPT	Overall Role
Pratiti Paul			✓	✓	✓	✓	Project Lead
Satvik Prasad			✓	✓			Analysis Lead
Aarush Gupta	✓	✓					ETL Lead
Krishiv		✓	✓				Data Lead
Utkarsh	✓					✓	Visualization Lead
Ravichandra Shinde					✓		ETL Lead